

A Per-Source SNR Labeling Pitfall in Open-Set Detection for Single-Channel Blind Source Separation

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Abstract—Evaluating out-of-distribution (OOD) detection for communication signals is routinely conditioned on the operating SNR. Because OOD score distributions drift with noise level, we show this conditioning is fragile to label–score misalignment: storing per-source SNR labels interleaved (`np.repeat`) while the scores they annotate are stacked slot-wise (`np.tile`) manufactures a compelling, spurious phenomenon in which two OOD scorer families appear complementary across SNR and a training-free SNR-routed ensemble appears to lift AUROC from 0.50 to 0.625 ± 0.031 , with apparent multi-seed significance, robustness, and graceful degradation. All of it is an artifact of the misalignment. We formalize the failure as an indexing-permutation confound and show it is general: *any* label–SNR decorrelation manufactures spurious per-SNR structure (routed weighted AUROC up to 0.77). We explain why the artifact survives exactly the checks practitioners trust, and report the corrected study under a deployment-faithful protocol: per-source detection of unknown modulations from separation embeddings is at chance across six scorers, five seeds, two backbones, and all tested conditions (routed 0.498 ± 0.020). Attribution experiments show the separator treats known and unknown sources alike and that a partial embedding margin present for interference-free sources (Mahalanobis AUROC 0.65) does not survive separation. An evaluation checklist is provided.

Index Terms—Blind source separation, open-set recognition, out-of-distribution detection, evaluation methodology, data leakage.

I. INTRODUCTION

SINGLE-CHANNEL blind source separation (SC-BSS) of co-frequency communication signals is increasingly paired with per-source modulation identification [1], [2], and a deployed receiver must additionally *reject* sources whose modulation it has never seen—an open-set recognition problem [3], [4] for which post-hoc OOD scorers of learned embeddings and logits [5], [6], [7], [8], [9] are the standard toolkit. Because OOD score distributions drift with noise, communication-signal evaluation is routinely conditioned on the operating SNR—per-SNR-bin profiles and weighted averages [14], [15], [4].

This letter shows that the conditioning step itself is a single point of failure. A one-line labeling-convention mismatch misaligns every known-pool score with the wrong SNR bin,

and because score scales drift with SNR, each “per-SNR bin” then compares known and unknown pools at *different* true SNRs: the bins measure SNR separability, not OOD separability. The artifact is dangerous precisely because it is *systematic*: it survives multi-seed averaging, significance testing, and robustness batteries—the safeguards practitioners trust [11], [12]. We contribute: (i) a formalization of the failure as an indexing-permutation confound and a synthetic-permutation study showing it is *general*—any label–SNR decorrelation, not only this bug, manufactures spurious per-SNR gains (routed weighted AUROC up to 0.765)—together with an explanation of why exactly the trusted checks fail to catch it; (ii) a deployment-faithful evaluation protocol (verified alignment, strict train/reference/test separation, unknown-free calibration, operating-point metrics); and (iii) the corrected study on an open-set SC-BSS benchmark: a systematic *negative* result with attribution, covering six scorers, five seeds, two backbones, channel and capacity interventions, leave-one-modulation-out (LOMO) splits, per-class near/far-OOD breakdowns, a calibrated single-threshold operating point, and an interference-free oracle probe.

II. OPEN-SET SC-BSS SETUP

Signal model and SNR definition. We study the two-source co-frequency mixture $y(t) = \alpha_1 s_1(t) + \alpha_2 s_2(t) + n(t)$ at complex baseband. Each s_i is a unit-power synthetic waveform (16 kHz sampling, 256 symbols at 16 samples/symbol, root-raised-cosine pulse shaping with roll-off 0.35, 3-tap complex Gaussian multipath normalized to unit gain, carriers 2000 Hz/2005 Hz with ± 5 Hz uniform jitter, $T = 4096$ samples [14]), mixed with $\alpha_1 = \alpha$, $\alpha_2 = 1 - \alpha$, $\alpha \sim U(0.4, 0.6)$, plus AWGN. Throughout, SNR denotes the *mixture* SNR $\text{SNR}_{\text{mix}} = 10 \log_{10}(\mathbb{E}|\alpha_1 s_1 + \alpha_2 s_2|^2 / \mathbb{E}|n|^2)$ —the quantity a blind receiver can estimate from y alone. Since the sources are unit-power, the per-source SNR of source i is $\text{SNR}_i = \text{SNR}_{\text{mix}} + 10 \log_{10}(\alpha_i^2 / (\alpha_1^2 + \alpha_2^2)) \approx \text{SNR}_{\text{mix}} - 3$ dB under near-equal mixing, and the SIR $20 \log_{10}(\alpha_1 / \alpha_2)$ sweeps $[-3.5, 3.5]$ dB with the α draw. The per-source label stored by the evaluation pipeline is this shared SNR_{mix} .

Modulations, protocols, pools. Modulations split into known $\mathcal{K} = \{\text{BPSK}, \text{QPSK}, \text{8PSK}, \text{16QAM}\}$ (training only) and held-out $\mathcal{U} = \{\text{64QAM}, \pi/4\text{-DQPSK}, \text{MSK}, \text{OFDM-QPSK}\}$; test protocols are kk/ku/uu by the number of unknown sources (deterministic seed 99999; 192 kk, 192 ku, and 96 uu mixtures per bin of the grid $\{-10, -5, \dots, 20\}$ dB). The *known pool*

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holds both slots of the *kk* mixtures and the *unknown pool* the OOD slot of each *ku* mixture plus both slots of *uu* (2688 sources each; 384 per SNR bin). The separator is a lightweight complex-valued CNN with complex squeeze-and-excitation (C-SE) attention ($\sim 69\text{K}$ parameters) plus a shared per-source modulation head (64-d embedding z , 4 logits f), trained with permutation-invariant SI-SDR + cross-entropy [16], [17] on known modulations only. Closed-set reference points (*kk*): SI-SDR -0.89 dB pooled, rising from -4.9 dB at -10 dB to $+0.6$ dB at 20 dB; modulation accuracy 0.407 —an interference-limited regime ($\text{SIR} \approx 0$ dB) in which separation is intrinsically hard; this regime is the stress test under which per-source OOD detection would have to operate.

Scorers and leakage control. Six post-hoc OOD scorers (larger = more OOD) are applied per source: Energy $-\log \sum_k e^{f_k}$ [7], MSP [5], ODIN [6], Mahalanobis [8], nearest-prototype distance, and a VOS-inspired (training-free, post-hoc) virtual-outlier distance [9]. All scorer hyperparameters are literature defaults fixed *a priori* (Energy/MSP temperature 1; ODIN $T=1000$, $\varepsilon=0.005$; Mahalanobis shared covariance, shrinkage 0.1; VOS-inspired scale 2, 100 outliers/class); every fitted quantity (prototypes, covariance, virtual outliers, thresholds) is estimated on a disjoint reference pool (seed 88888, *kk* protocol), and **no unknown-class sample is used for any fitting, selection, or calibration step**—the routing boundary (0 dB) is fixed *a priori*, and the LOMO study selects boundaries exclusively on a further pseudo-ODD validation split (seed 77777). The rejection threshold τ is the 95th percentile of known-pool reference scores (5% false-rejection target). The *a-priori SNR-routed* score uses Energy at $\text{SNR} \leq 0$ dB and Prototype otherwise.

III. THE PITFALL

Mechanism and formalization. Index the test mixtures $m = 1, \dots, N$ in SNR-block order. The evaluation stores per-source scores q_j stacked *tile-wise* and per-source SNR labels ℓ_j written *interleaved* (`np.repeat`), so score j belongs to mixture $\sigma_q(j) = ((j-1) \bmod N) + 1$ but is paired with the label of mixture $\sigma_\ell(j) = \lceil j/2 \rceil$: an indexing-permutation mismatch $\sigma_q \neq \sigma_\ell$ (Fig. 2). Let $c(m)$ be the true SNR of mixture m and suppose the score distribution drifts with SNR, $p(s | c) \neq p(s | c')$. Binning by ℓ assigns to stored bin b scores whose *true* SNRs are $\{c(\sigma_q(j)) : c(\sigma_\ell(j)) = b\} \neq \{b\}$ wherever the permutations disagree; the unknown pool, whose labels were exact, is unaffected. Each per-bin AUROC then estimates $P(s_u > s_k | c_u = b, c_k \in C_b)$ with $C_b \neq \{b\}$: it measures the score’s *SNR separability*, not OOD separability, with a magnitude set by the deterministic label-truth SNR gap (up to 15 dB here). A one-line audit exposes the corruption: binning the known pool’s stored *modulation* labels by the stored SNR labels gives per-bin class counts 114/96/100/74, impossible for a protocol contributing exactly 96 mixtures per known modulation per true bin; the corrected (tile) labels recover 96/96/96/96.

Generality: any label-SNR decorrelation manufactures the phenomenon. The repeat-vs-tile bug is one member of a class. Re-binning the *corrected* scores with synthetic label

permutations π over the known pool (5-seed mean weighted AUROC of the *a-priori* routed score): $\pi_{\text{identity}} = 0.498$; π_{repeat} (the original bug) = 0.625; $\pi_{\text{reverse}} = 0.765$; ten uniform *random* permutations = 0.641 ± 0.005 ; block-structure preserving shifts are exact no-ops (0.498). Per-bin AUROCs reach 0.85–0.87 under π_{reverse} . Two confounds combine: mismatched known bins separate by their true-SNR composition (under π_{random} each bin approximates “full known pool vs unknown at b ”, exposing monotone SNR gradients—Energy’s mismatched AUROC decreases with b while Prototype’s increases), and the fixed *a-priori* routing rule harvests both gradients by construction. Notably, single-scorer weighted averages stay near 0.48–0.51 under every π : the average can look innocent while the per-bin structure is entirely spurious—which is exactly how the artifact presented as *complementarity* rather than as a global lift. Label-index alignment is therefore not an implementation detail but a confound-control condition for conditioned evaluation.

The manufactured phenomenon. Under the buggy labels the benchmark presented a perfect complementarity (Prototype 0.84 AUROC at 10 dB, collapsing below 0 dB; Energy 0.74 at -5 dB, falling to 0.22 at 10 dB), and routing on it appeared to lift the pair-weighted AUROC from 0.50 to 0.625 ± 0.031 (five seeds) against a post-hoc oracle of 0.681 (Fig. 1a). It also produced a complete supporting narrative: “graceful degradation” under simulated SNR-estimation error ($\sigma = 1/3/6$ dB $\rightarrow 0.597/0.593/0.576$), “robustness” to SIR/timing/carrier perturbations (0.58–0.66), and “consistent” ablations (center loss 0.568, carrier gaps 0.615–0.633, embedding dims 0.588–0.631).

Why it survives the standard checks. The misalignment is a deterministic function of the data layout, identical for every seed and condition: multi-seed averaging *confirms* it (± 0.031), and significance tests are significant precisely because the SNR-mismatch signal is real—what is wrong is the bins’ interpretation. Robustness perturbations act on the signals, not the label geometry, so each pass reads as independent evidence; the σ -sweep’s graceful decay is exactly what the artifact predicts, since noising the SNR merely blurs the mismatch signal. And the pattern is mechanistically narratable (“confidence inverts at high SNR”), which is how it survives internal scrutiny. The discriminating check is the one nobody thinks to run: verifying that the *labels* are aligned.

How it was caught. Replacing the oracle SNR in the router with a real blind SNR estimator (median-eigenvalue subspace estimator, validated at 0.51 dB overall RMSE; the classical M2M4 is inadequate for this signal class [13]) required pairing waveform-derived estimates with stored scores sample by sample. The alignment verification written for that purpose—regenerating the deterministic test set and checking stored arrays against the data order—failed immediately and exposed the repeat-vs-tile mismatch.

IV. CORRECTED PROTOCOL AND RESULTS

Protocol. (i) **Verified label alignment:** every dumped label array is checked against regenerated data (sequence equality, per-bin counts, per-bin label multisets). (ii) **Strict separation:**

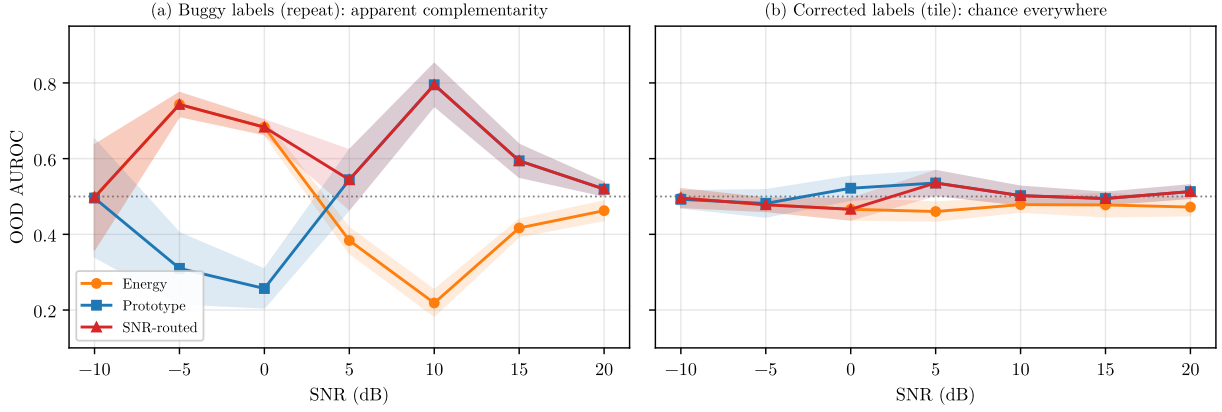


Fig. 1. Per-SNR OOD AUROC of the *same* score dumps (5-seed mean $\pm$ std) under (a) the buggy repeat labels and (b) the corrected tile labels. The apparent complementarity—Energy peaking at low SNR, Prototype at high SNR, the routed rule following the better branch—is manufactured by the label misalignment; corrected, every profile is flat at chance.

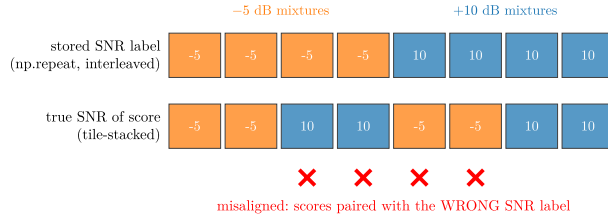


Fig. 2. The misalignment. Scores are stacked tile-wise (all slot-1 entries, then all slot-2), so their true SNR sequence alternates blocks; the stored labels repeat each mixture’s SNR interleaved. Entries past the slot boundary (red crosses) are paired with the wrong SNR—by up to 15 dB in the real seven-block test layout.

training, the reference pool (seed 88888), and the test set (seed 99999) never mix; all fitted OOD quantities come from the reference pool. (iii) **Unknown-free calibration:** τ is the 95th percentile of known-pool reference scores; no unknown-class data is used for any fitted quantity, and the routing boundary is fixed a priori. (iv) **Operating-point metrics**—Det@ τ , FPR95, OSCR [10], joint known/unknown accuracy—alongside per-SNR and pooled AUROC. Seed standard deviations describe training-run variability; we additionally report 95% bootstrap confidence intervals over test samples—the two uncertainties are distinct.

Corrected results. Under verified labels every scorer is flat at chance at every SNR (Fig. 1b; Table I): the routed ensemble reaches 0.498 ± 0.020 , and the post-hoc oracle over all six scorers 0.549 ± 0.032 , marginally above chance by per-bin max-selection alone. MSP sits consistently *below* chance (0.43)—the known-class head is confidently wrong on unknown modulations; Mahalanobis sits marginally above (0.520 ± 0.050 at seed level, though its concatenated-sample CI excludes 0.5), driven by far-OOD classes and high-SNR bins (below). Nothing survives perturbation because nothing was there: the SNR-estimator-noise sweep is flat (0.501 – 0.502), SIR/timing/CFO conditions give 0.487 – 0.494 , carrier gaps 10–500 Hz give 0.496 – 0.503 , center loss 0.487 , embedding dims 16–128 give 0.477 – 0.508 . LOMO splits (routing boundary

TABLE I
WEIGHTED-AVERAGE PER-SNR AUROC, ARTIFACT VS CORRECTED LABELS (5 SEEDS, MEAN $\pm$ SEED STD), WITH CORRECTED POOLED AUROC AND ITS 95% BOOTSTRAP CI OVER THE CONCATENATED TEST SAMPLES. THE ARTIFACT COLUMN IS REPRODUCED FROM THE ARCHIVED MISLABELED DUMPS FOR DOCUMENTATION ONLY. UNDER CORRECTED LABELS THE POST-HOC ORACLE EXCEEDS THE A-PRIORI ROUTED SCORE BY 0.053 (PAIRED BOOTSTRAP 95% CI [0.043, 0.064])—A SIGNIFICANT GAP THAT IS PURELY PER-BIN MAX-SELECTION AT CHANCE.

Scorer	Artifact wavg	Corrected wavg	Corrected pooled [95% CI]
Energy	0.487 ± 0.032	0.476 ± 0.019	0.482 [0.477, 0.491]
MSP	0.435 ± 0.012	0.431 ± 0.009	0.433 [0.429, 0.443]
ODIN	0.495 ± 0.030	0.480 ± 0.019	0.486 [0.484, 0.498]
Mahalanobis	0.526 ± 0.043	0.520 ± 0.050	0.524 [0.509, 0.523]
Prototype	0.502 ± 0.008	0.506 ± 0.010	0.509 [0.494, 0.507]
VOS-inspired	0.502 ± 0.008	0.506 ± 0.010	0.509 [0.494, 0.507]
SNR-routed	0.625 ± 0.031	0.498 ± 0.020	—
Oracle (6, post-hoc)	0.681 ± 0.039	0.549 ± 0.032	—

selected exclusively on pseudo-OOD validation data) stay at 0.482 – 0.509 , and the selected boundary is unstable (9/12 runs pick 0 dB; 3/12 find no crossover and fall back to all-Prototype)—there is no crossover to find. A second backbone without the C-SE blocks (weaker closed-set model) behaves identically: routed 0.508 ± 0.011 , no complementarity.

Operating point. At the calibrated single threshold (Table II) the detector rejects 5.8% of unknown sources against a 5.2% false-rejection rate—a near-useless trade that FPR95 (0.954), OSCR (0.196) and joint accuracy (0.163) confirm. Per SNR, Det@ τ is *exactly* 0 at ≤ 0 dB and rises to 0.14 at 20 dB, with the false-rejection rate tracking it ($0 \rightarrow 0.139$): the global threshold tracks the known pool’s SNR-driven score drift and the unknown pool drifts identically. The failure is distributional, not a threshold-choice artifact—no operating point exists because the score distributions do not separate.

Near vs far OOD. Breaking the unknown pool down by class (weighted-average AUROC, corrected): for embedding-distance scorers the far-OOD families score weakly higher (Mahalanobis: MSK 0.556 ± 0.041 , OFDM-QPSK 0.551 ± 0.117 ; Prototype: $0.500/0.527$) than the near-OOD ones (64QAM $0.478/0.478$, $\pi/4$ -DQPSK $0.496/0.516$); for logit-

TABLE II

OPERATING-POINT METRICS OF THE ROUTED SCORE AT THE CALIBRATED 5%-FRR THRESHOLD τ (5 SEEDS, MEAN $\pm$ STD). VARIANT A: SPLIT-HALF REFERENCE; VARIANT B: DISJOINT REFERENCE POOL (SEED 88888) WITH REFITTED PROTOTYPES.

Metric	Variant A	Variant B
Det@ τ (unknown detection)	0.058 ± 0.035	0.058 ± 0.038
False-rejection rate (target 0.05)	0.052 ± 0.001	0.051 ± 0.003
FPR95	0.954 ± 0.009	0.954 ± 0.010
OSCR [10]	0.196 ± 0.013	0.197 ± 0.012
Joint known/unknown accuracy	0.163 ± 0.030	0.217 ± 0.028

based scorers the pattern *inverts*—MSP on OFDM-QPSK is 0.359 ± 0.027 , i.e. the multicarrier waveform looks *more* indistribution to the known-class head than the known pool itself. Neither direction is deployable; the near/far intuition does not transfer to post-hoc scoring of separated sources.

Attribution. Three controlled observations bound the explanation. (1) Separation quality is class-independent: SI-SDR of known vs unknown sources is -0.89 vs -1.00 dB pooled (within 0.21 dB at every SNR bin)—the separator does not distort unknown modulations more than known ones. (2) A partial margin exists without interference: applied to clean single-source inputs (a probe, not a deployed configuration), embedding-geometry scorers reach Mahalanobis AUROC 0.660 ± 0.046 —rising from 0.52 at -10 dB to 0.77 at 10 dB, and 0.82 on the far-OOD OFDM waveform—while logit scorers remain at or below chance (0.42–0.44) even there. (3) Under the two-source separated condition every scorer falls to ≈ 0.5 (Table I). Together, the observed results suggest that the separation and known-class objectives do not provide a sufficiently exploitable known/unknown margin for post-hoc scorers *operating on separated sources*: a margin that partially exists in the embedding does not survive the mixture/separation condition (residual interference at SIR ≈ 0 dB), and the logit pathway never carries one. Scope: these conclusions hold for the evaluated synthetic benchmark and protocol; dedicated open-set modulation recognition [4] with unknown-aware training is a separate, harder problem.

V. CONCLUSION

A repeat-vs-tile labeling mismatch manufactured an entire SNR-conditioned phenomenon that survived multi-seed averaging, significance intuition, and every robustness check; a synthetic permutation study shows the failure is a general indexing confound, not a one-off bug. Corrected, and under a deployment-faithful protocol, post-hoc OOD detection of unknown modulations from separation embeddings remains at chance on the evaluated synthetic SC-BSS benchmark. For conditioned OOD evaluation we recommend: (1) *align labels against regenerated data* before trusting any conditioned metric (sequence, per-bin counts, label multisets)—no averaging or robustness battery substitutes for this check; (2) *multi-seed averaging is not a pipeline check*—it averages random error and confirms systematic error; (3) *separate train/reference/test strictly* and calibrate operating points without unknown-class data, reporting Det@ τ alongside AUROC; (4) *treat per-bin oracle*

bounds as bounds, never as deployable headlines. Code, generators, and both the corrected and archived (artifact) dumps are released at https://github.com/BinNong/single_channel_source_seperation/tree/main/paper3_open_set (pinned release tag paper3-spl).

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